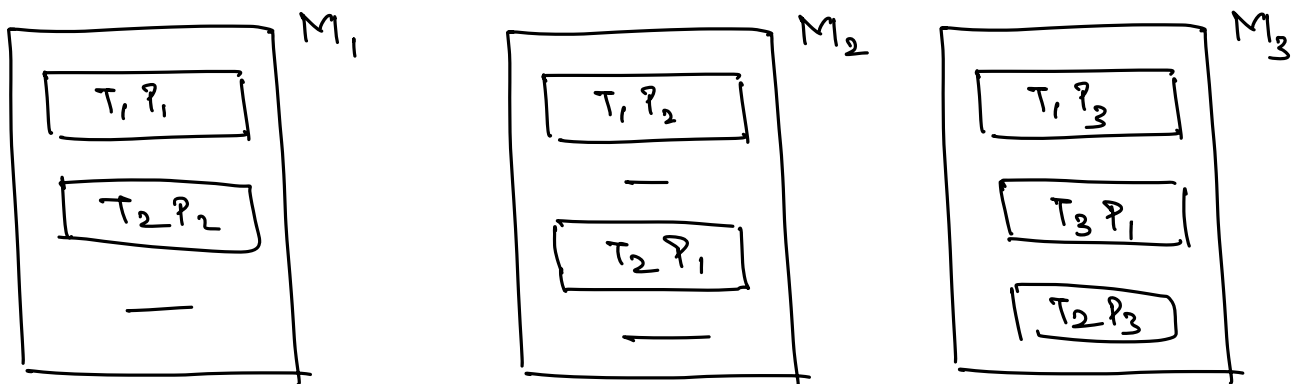
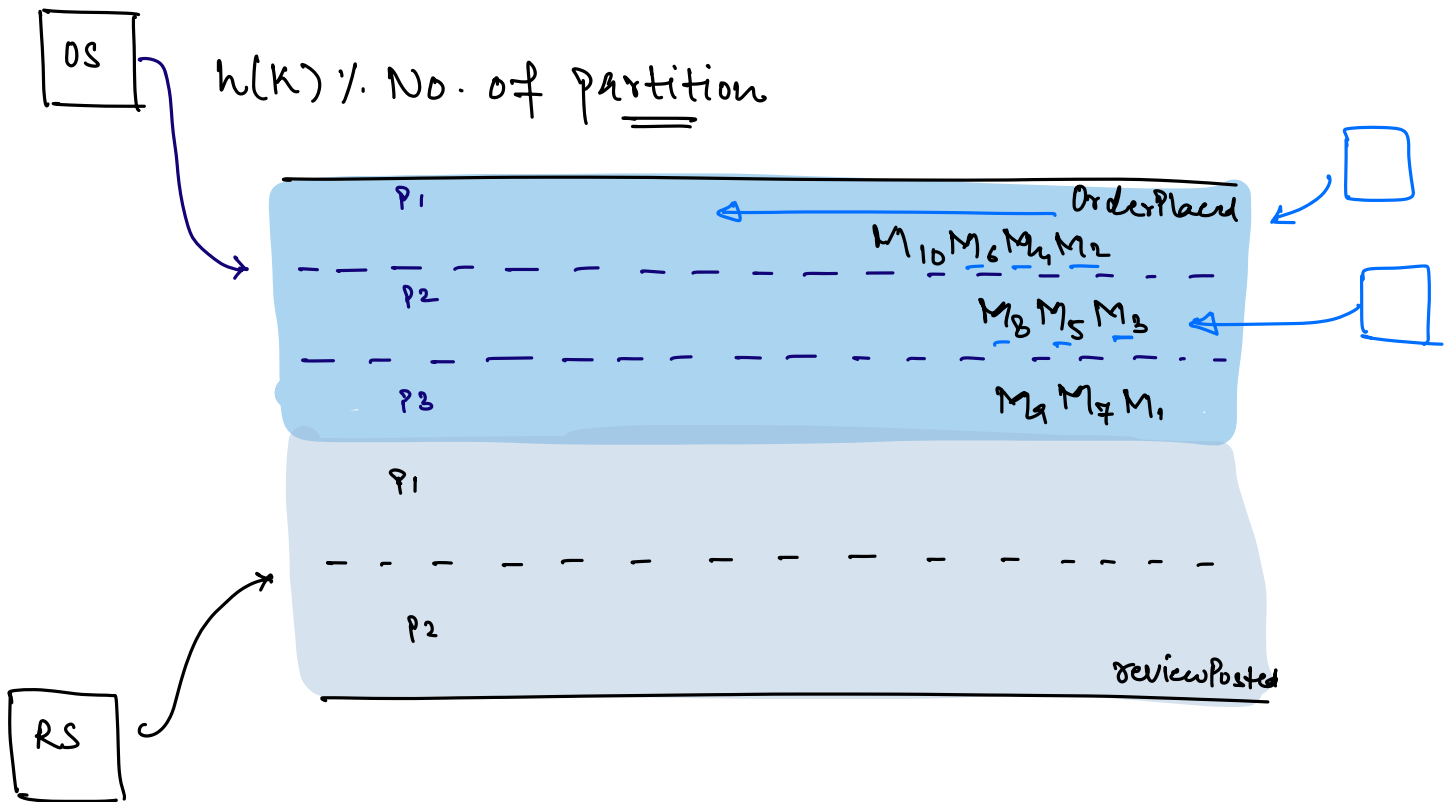
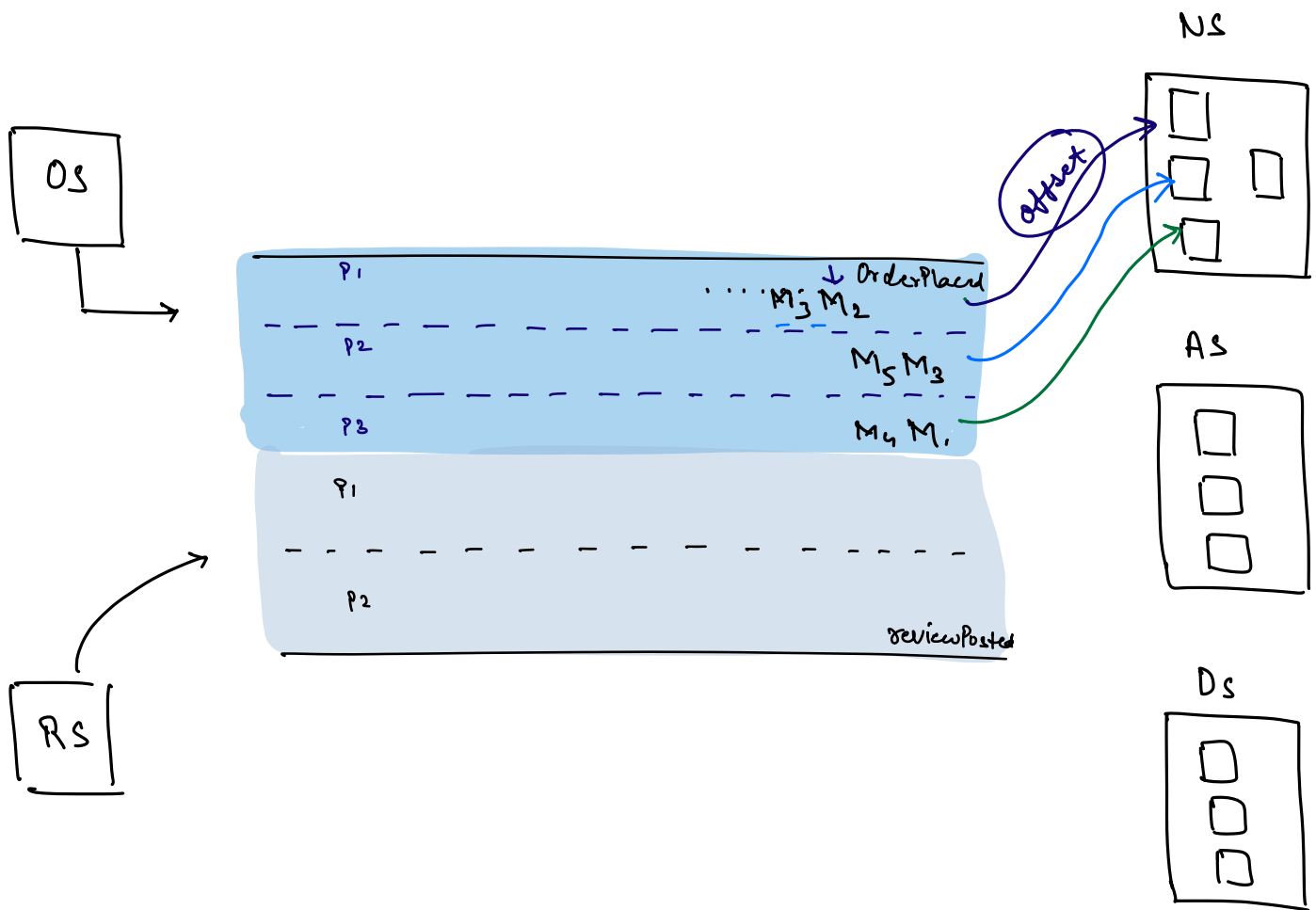


Agenda

→ Kafka Continued.



⇒ Kafka : FIFO → within the partition



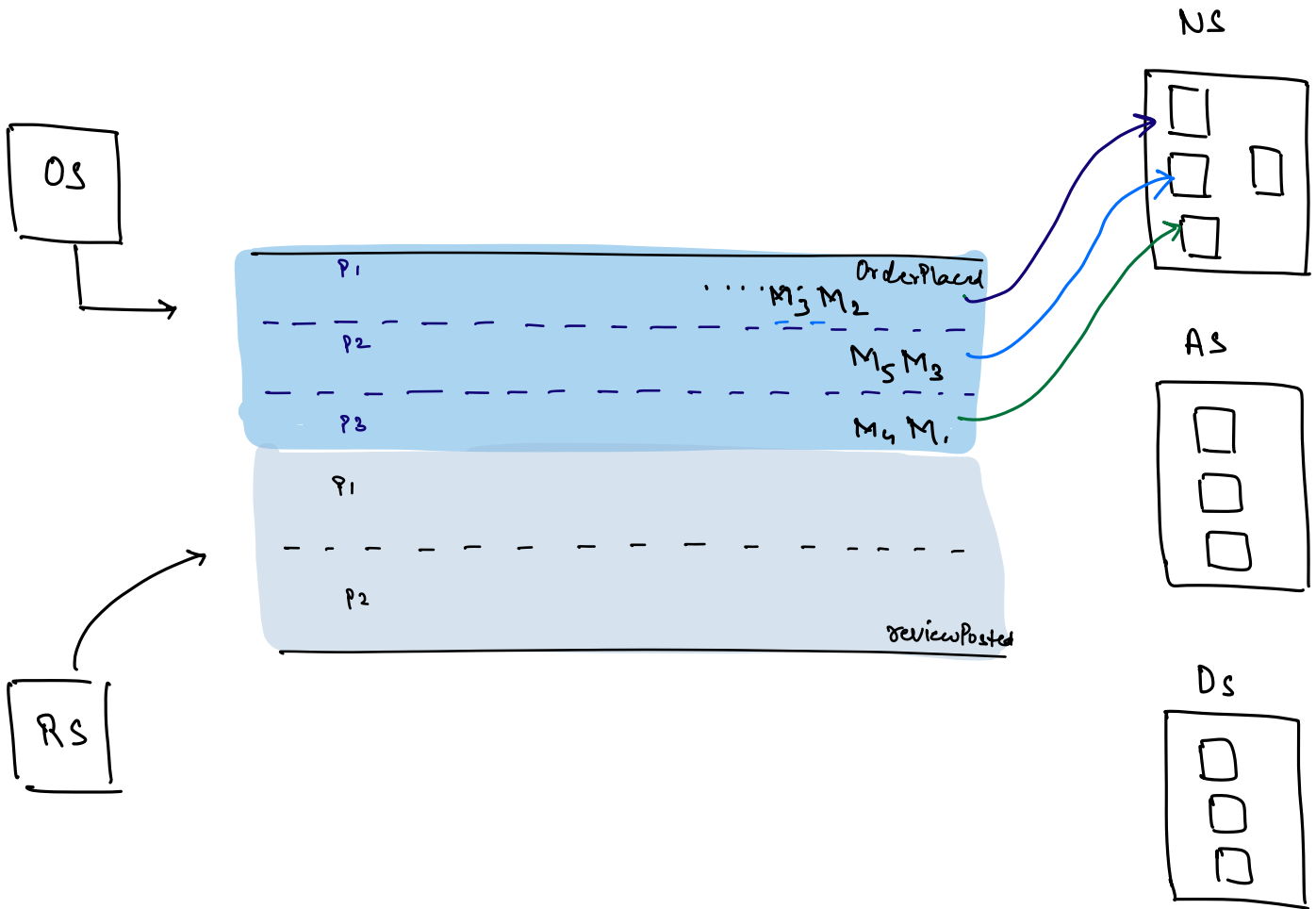
If # of consumers in a group > # of partitions in a particular topic

→ Some of the consumers will sit idle.

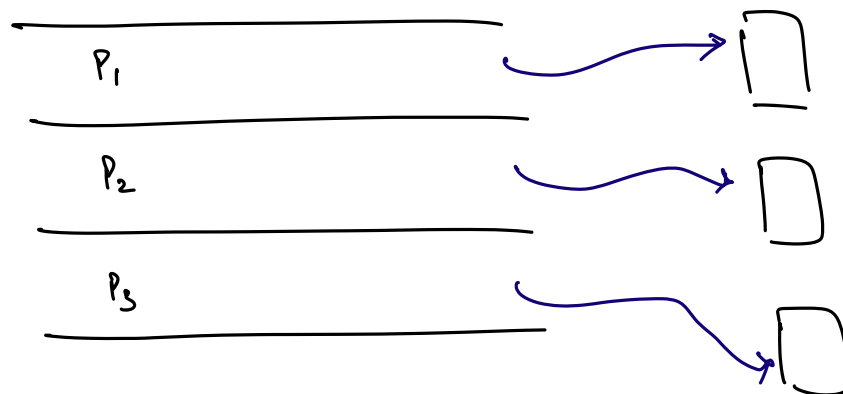
If # of consumers $=$ # of partitions
in a group in a particular topic



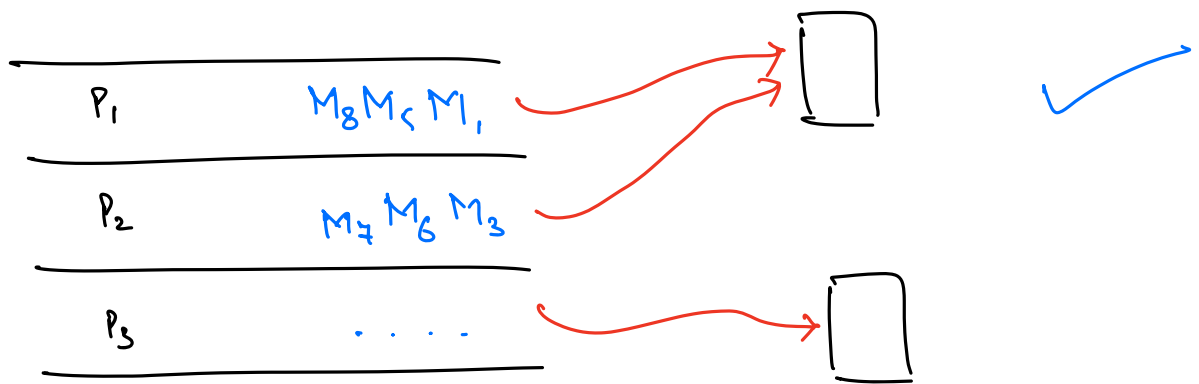
If # of consumers $<$ # of partitions
in a group in a particular topic



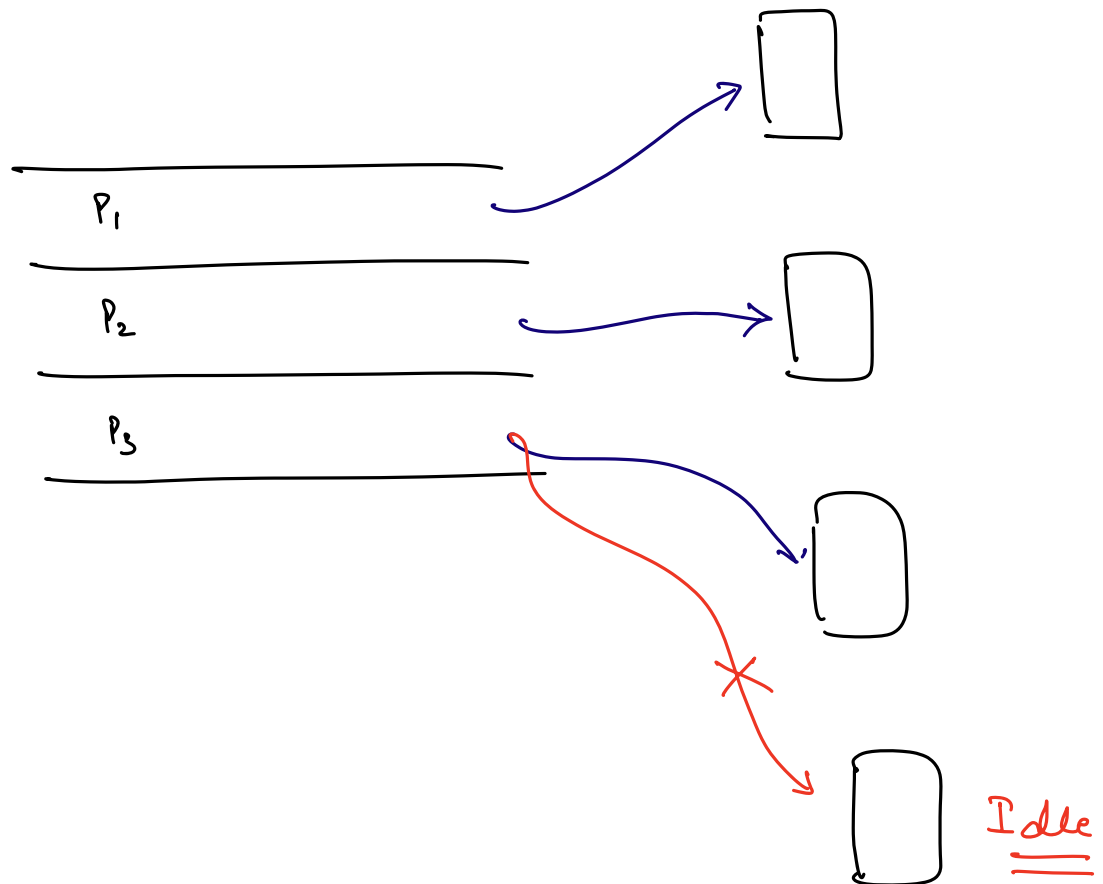
1)



2)

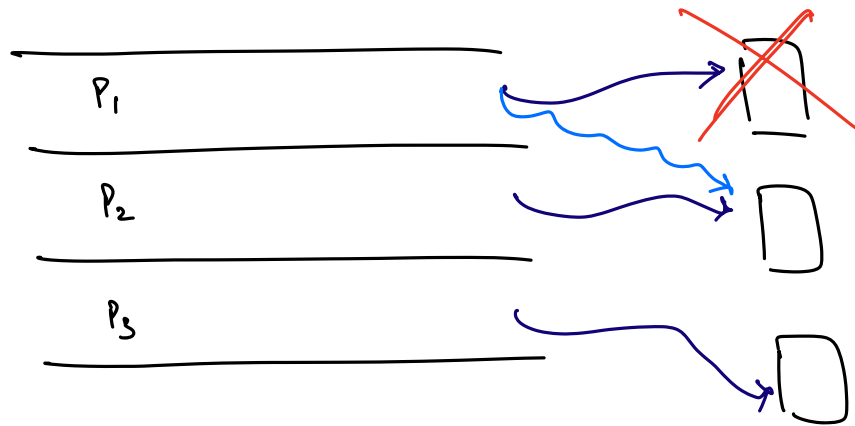


3)



of partitions $\geq$ # of consumers.

⇒ we always create more # of partitions proemptively.



⇒ If a consumer goes down:

- 1) Assigning a new consumer.
- 2) kafka reorganizes partitions to the remaining consumers.

Delivery Guarantee in Kauka.

→ At most Once : No duplicate delivery.

↘ 0 or Once.

↓
lost.

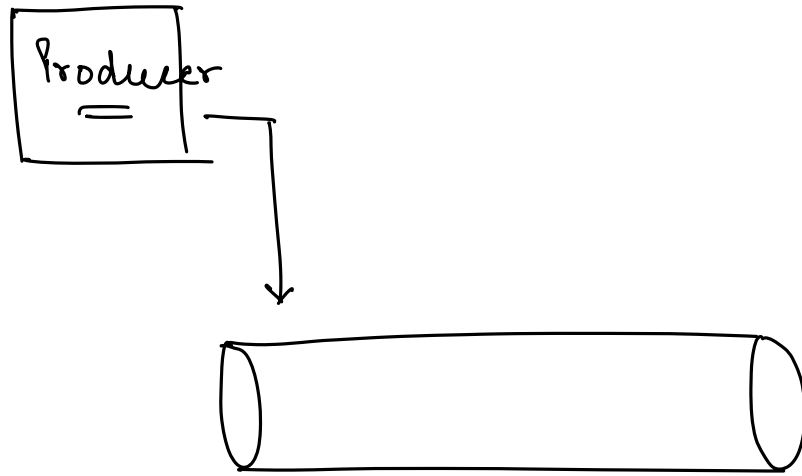
→ At least Once : 1 or more time

↳ duplicate delivery.

→ Exactly Once.

↳ No message lost & No duplicate delivery.

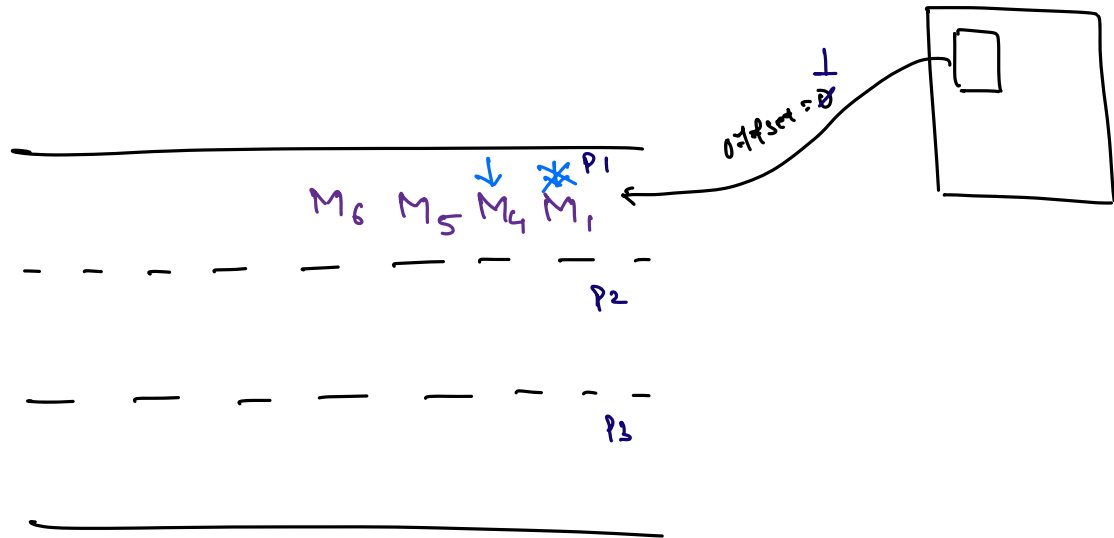
What if a Producer | Kafka Broker Crashes.?



Kafka producer API supports Idempotency.
(optionally).

Kafka ensures Durability

Consumer.



⇒ Consumer commits the offset to Kafka.

I) Reads the Event → Commit Offset → Process the Event.

⇒ At most Once delivery.

II) Reads the Event → Process the Event → Commit Offset



At Least Once delivery.

⇒ Exactly Once.

↓
Kafka Streams.

Sharding & Replication.

↳ Kafka Brokers.

SHARDING. : Messages in Kafka are sharded by topic & partition.

REPLICATION. : Each topic's partition is replicated across multiple brokers in a master-slave configuration with Quorum.

